

packages/graph-core/src/search.ts

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Overview

A new `search.ts` module is added under `packages/graph-core/src`. It introduces a lightweight full-text search layer powered by **Fuse.js** for `GraphNode` objects.

Key changes

- **Imports** (R1-R2): `Fuse` and `IFuseOptions` from `*fuse.js`
- and `GraphNode` from `./types.js`.
- **Interfaces** (R4-R7, R9-R12):
 - `SearchResult` exposes `nodeId` and `score` (0=perfect match, 1=worst).
 - `SearchOptions` allows optional `types` filtering and a `limit`.
- **Fuse configuration** (R14-R25): `FUSE_OPTIONS` sets key weights, threshold, and enables extended search.
- **SearchEngine class** (R27-R65):
 - Stores `nodes` and a `Fuse` instance.
 - `search(query, options?)` trims the query, joins tokens with `|` for OR matching, runs Fuse, filters by `types`, limits results, and maps to `SearchResult`.
 - `updateNodes(nodes)` rebuilds the internal index.

Impact

- **Functionality:** Provides a public search API for graph nodes, enabling UI search bars and tooling.
- **Compatibility:** Requires adding `fuse.js` to dependencies; no breaking changes to existing modules.
- **Observability:** No new logs; results are deterministic based on Fuse scoring.

Risks & follow-ups

- **Dependency version:** Ensure `fuse.js` is pinned to a compatible major version.
- **Score interpretation:** `score` is inverted (0=perfect); confirm consumers handle this convention.
- **Type safety:** `GraphNode["type"]` filtering assumes `type` exists; verify all node shapes meet this contract.
- **Performance regression:** unknown from the available diff/scan evidence; benchmark search on the largest graph to confirm acceptable latency and memory usage.